Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 32: Approved Solutions

The Principle of Inclusion-Exclusion

Covers Chartrand & Zhang, Section 8.2

Status of this set

Ungraded practice. Solutions released with it. Show the full expression, not just the final number.

Part I: How to Practice This Material

The two formulas

$$|A \cup B| = |A| + |B| - |A \cap B|$$

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

Signs alternate so that every Venn region is counted exactly once.

Two habits

Count the complement when the direct count is messy: $|\overline{A}| = |U| - |A|$.
For divisibility, $|A_a \cap A_b|$ counts multiples of the least common multiple — which is ab only when a and b share no factor.

Three mistakes to expect

1. **Omitting $+|A \cap B \cap C|$.** The answer comes out too small.
2. **Using ab for the pairwise intersection** when a and b share a factor.
3. **Adding when the sets overlap.** That is what this principle exists to correct.

Part II: Practice Problems

Section A — Two sets

1. How many integers from 1 to 1000 are divisible by 4 or by 6?

Approved solution

$|A_4| = \lfloor 1000/4 \rfloor = 250$ and $|A_6| = \lfloor 1000/6 \rfloor = 166$.
An integer divisible by both 4 and 6 is divisible by $\operatorname{lcm}(4,6) = 12$, not 24. So $|A_4 \cap A_6| = \lfloor 1000/12 \rfloor = 83$.

$$|A_4 \cup A_6| = 250 + 166 - 83 = 333.$$

This is the least-common-multiple trap. Using 24 would give $|A_4 \cap A_6| = 41$ and an answer of 375, which is wrong. The product works only for numbers sharing no factor.

2. How many bit strings of length 8 begin with 11 or end with 00?

Approved solution

Begin with 11: the other 6 bits are free, so $2^6 = 64$.
End with 00: likewise $2^6 = 64$.
Both: first two and last two fixed, 4 bits free, so $2^4 = 16$.

$$64 + 64 - 16 = 112.$$

Section B — Three sets

3. How many integers from 1 to 200 are divisible by 2, 3, or 7?

Approved solution

$$|A_2| = 100, \quad |A_3| = 66, \quad |A_7| = 28,$$
$$|A_6| = 33, \quad |A_{14}| = 14, \quad |A_{21}| = 9, \quad |A_{42}| = 4.$$
$$|A_2 \cup A_3 \cup A_7| = 100 + 66 + 28 - 33 - 14 - 9 + 4 = 142.$$

Since 2, 3, and 7 are pairwise coprime, each intersection is counted by the product: 6, 14, 21, and 42.

4. A survey of 100 moviegoers found 60 buy popcorn, 45 buy candy, 55 buy a soft drink, 25 buy popcorn and candy, 30 buy popcorn and a drink, 20 buy candy and a drink, and 10 buy all three.

(a) How many buy at least one of the three?

(b) How many buy none of them?

Approved solution

(a) $60 + 45 + 55 - 25 - 30 - 20 + 10 = 95$.
(b) $100 - 95 = 5$ — complement counting.

Check your work with a sanity test: the answer to (a) can never exceed the total surveyed. If inclusion–exclusion ever gives you more than 100 here, you have dropped a subtraction.

Section C — Complement counting

5. How many integers from 1 to 300 are divisible by none of 2, 3, 5?

Approved solution

Count the ones that *are* divisible by at least one:

$$150 + 100 + 60 - 50 - 30 - 20 + 10 = 220.$$

So the answer is $300 - 220 = 80$.

Counting these directly would require checking each integer against three conditions. Counting the complement turns it into one application of a formula — which is the whole point of the technique.

Section D — Writing

6. In four to six complete sentences, explain why the three-set formula alternates signs. Track a single element that lies in all three sets, and say how many times each term counts it.

Approved solution

Model answer. Consider an element belonging to all three of A, B, and C. The first three terms count it once each, for a total of three. The three pairwise intersections each contain it, so subtracting them removes it three times, bringing the running total to zero. Adding the triple intersection puts it back exactly once, which is correct. The same bookkeeping works for an element in exactly two of the sets: it is added twice, subtracted once by the one pairwise term containing it, and is not in the triple intersection, so it is counted once. Every Venn region ends up counted exactly once, and that is what makes the alternating signs necessary rather than arbitrary.

Full credit requires tracking at least the all-three element through all four stages. The two-set case is a bonus.

Instructor notes

- **Problem 1 is the lcm trap** and the most common error in this section. Using 24 instead of 12 gives 375 — plausible-looking and wrong.
- **Problem 6** is the conceptual item. Students who can track one element through all four terms understand why the signs alternate; those who memorised the formula cannot.
- **Problem 5** shows complement counting doing real work.
- Reserved for Homework 12: Section 8.2 even-numbered exercises, including Exercise 16 (four sets).